INCH-POUND

MIL-DTL-32362
7 December 2010
SUPERSEDING
MS51303B
14 September 1973

DETAIL SPECIFICATION

LIGHT, MARKER, CLEARANCE-BLACK

Inactive for new design after 7 December 2010. For new design, use MIL-PRF-32213.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

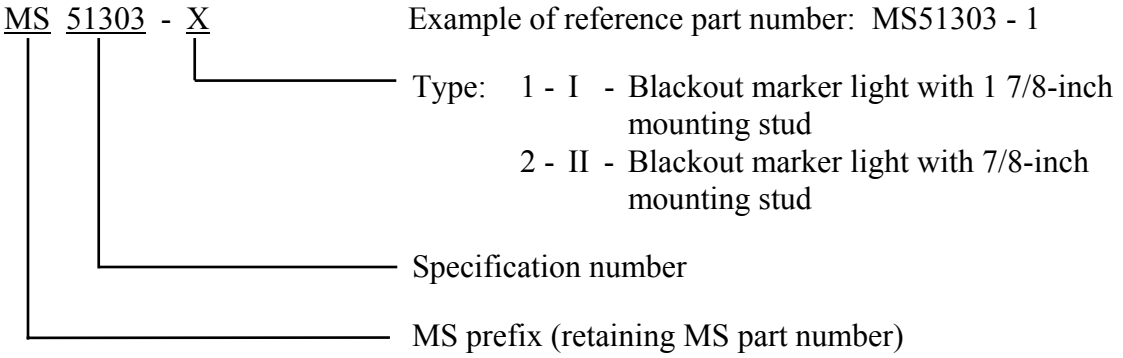
1.1 Scope. This specification covers two types of vehicular blackout marker lights.

1.2 Classification. The blackout marker lights are of the following types as specified below (see 6.2).

1.2.1 Types. The types of blackout marker lights are as follows:

Type I - Blackout marker light with 1 7/8-inch mounting stud
Type II - Blackout marker light with 7/8-inch mounting stud

1.3 Part or identifying number (PIN). The PIN to be used for blackout marker lights acquired to this specification is created as follows:



Comments, suggestions, or questions on this document should be addressed to Defense Logistics Agency Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.daps.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARD

FED-STD-H28	- Screw-Thread Standards for Federal Services

COMMERCIAL ITEM DESCRIPTION

A-A-52463	- Lamp Bulbs, Incandescent, Miniature, Single and Double Contact, Bayonet Candelabra Base

DEPARTMENT OF DEFENSE SPECIFICATION

MS33800	- Connector, Receptacle, Electrical, Pin Contact No. 8, 12, 14 and 16 AWG, Waterproof

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-130	- Identification Marking of U.S. Military Property

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	- Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at http://www.asq.org or from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53023.)

ASTM INTERNATIONAL

ASTM B85/B85M	- Standard Specification for Aluminum-Alloy Die Castings
ASTM B117	- Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM G21	- Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

(Copies of these documents are available online at http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The blackout marker light shall be as shown in figure 1.

3.2 Design.

3.2.1 Wiring diagram and lamp. The blackout marker light does not have a wiring diagram since it only has one 24 volts direct current (VDC) terminal connection. The circuit is considered completed once the blackout marker light has been installed in the intended application. The mounting stud is installed into a metal boss that is electrically connected to the application's vehicle frame, which provides for the grounding connection. The lamp used in the blackout marker light is AA52463-A08, as specified by A-A-52463.

3.2.2 Visibility. The blackout marker light shall have a maximum visibility of not less than 1000 ±200 feet on a dark, cloudy night with no moon with 24 VDC volts direct current applied to the bulb. The two white apertures shall appear to the unaided eye to be substantially equal in brightness.

3.2.3 Apertures. The two apertures will appear as one from a distance of 60 feet or more.

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3.2.4 Waterproofness. The blackout marker light shall display no evidence of leakage under visual examination and shall be operable, in accordance with 3.4, during and subsequent to submersion when subjected to the tests of 4.5.1.

3.2.5 Corrosion resistance. The blackout marker light shall display no evidence of corrosion under visual examination and shall be fully operational, in accordance with 3.4, following the salt atmosphere test of 4.5.2.

3.2.6 Fungus resistance. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicide agent in accordance with ASTM G21. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.2.7 Identification. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. Identification shall include the PIN (see 1.3) and the manufacturer's Commercial and Government Entity (CAGE) code.

3.3 Materials.

3.3.1 Housing. The main two-piece housing shall be constructed from aluminum alloy A360 in accordance with ASTM B85/B85M.

3.3.2 Threads. All screw threads shall be in accordance with FED-STD-H28.

3.3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life-cycle costs.

3.4 Operation. The blackout marker lights shall give no indication of electrical or mechanical faults when tested in accordance with 4.4.3.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirement specified herein is classified as conformance inspection (see 4.3).

4.2 Sampling.

4.2.1 Lot. For purposes of sampling, a lot shall consist of not more than 100 blackout marker lights of the same type, assembled from the same batch of components or subassemblies, and offered for delivery at one time.

4.2.2 Sampling for conformance inspection. A random sample of blackout marker lights shall be selected from each lot and subjected to the inspections specified in 4.3. Sampling shall

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be in accordance with ASQ Z1.4, inspection level III, with the acceptance quality limit (AQL) as specified in the contract (see 6.2).

4.3 Conformance inspection. Unless otherwise specified (see 6.2), each of the sample blackout marker lights selected in accordance with 4.2.2 shall be subjected to the examinations and tests specified in table I.

TABLE I. Conformance inspection.

Inspection	Requirement	Verification	Examination	Test
Design and materials	3.2 through 3.3.3	4.4.2, 4.4.4, 4.4.5, 4.5.1, 4.5.2	X	-
Operation	3.4	4.4.3	-	X

4.4 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously approved or previously qualified designs.

4.4.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance. Refer to contract (see 6.2) for alternatives to verifications required by this specification.

4.4.2 Design and materials. Conformance to 3.2 through 3.3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Visual verification discovering that the required components specified herein are not used shall constitute failure of this test (see 4.4).

4.4.3 Operation. Each sample blackout marker light shall be visually tested by energizing the blackout marker light with 24 VDC to verify proper operation and illumination. Failed lamps may be replaced.

4.4.4 Visibility. Each sample blackout marker light shall be visually tested by energizing the blackout marker light with 24 VDC to verify the maximum visibility, in feet, on a dark, cloudy night with no moon. The two white apertures shall appear with the unaided eye to be substantially equal in brightness.

4.4.5 Apertures. Each sample blackout marker light shall be visually tested by energizing the blackout marker light with 24 VDC to measure in feet at what distance the two apertures appear as one.

4.5 Tests.

4.5.1 Waterproofness. The blackout marker light shall withstand the waterproofness tests of 4.5.1.1 through 4.5.1.4.

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4.5.1.1 Saline solution. The salt used shall be sodium chloride containing on a dry basis not greater than 0.1 percent of sodium iodide and not greater than 0.2 percent of total impurities. The solution shall be prepared by dissolving 5 parts by weight of salt in 95 parts by weight of distilled water or other water containing not greater than 200 parts per million of total solids. The solution shall be kept free of sediment by filtration or decantation.

4.5.1.2 Preparation and initial testing. The blackout marker light, with its electrical connections, shall be submerged in a container with the uppermost surface of the assembly a minimum of 1 inch below the surface of the saline solution. The container and assembly shall be installed in a transparent, sealed chamber capable of supporting both vacuum and pressurization. The assembly shall initially be operated while submerged for 30 minutes and shall be carefully observed during its entire period of submersion.

4.5.1.3 Vacuum submersion testing. The chamber shall be evacuated to a pressure 6 pounds per square inch (psi) below atmospheric so as to apply a minimum of 6 psi in internal pressure to all voids within the assembly. During this period, the blackout marker light shall be carefully observed for leakage, as evidenced by bubbles escaping from the interior of the assembly. Any leakage shall be considered as noncompliance with the waterproof requirement and the assembly shall be rejected. Bubbles that are the result of entrapped air on the exterior surfaces of the component shall not be considered leakage.

4.5.1.4 Pressure submersion testing. The chamber shall then be pressurized to 6 pounds above atmospheric and the assembly again operated for 30 minutes. The assembly shall then be disassembled as normally required in servicing, and inspection made for the presence of water. If water is present, the assembly shall be rejected. If the assembly is dry, it shall be reassembled and subjected to 15 hours of dry operation (3 separate 5-hour periods). Insulation breakdown or other damage that would impair mechanical or electrical operation of the assembly shall be considered as evidence of failure and the assembly shall be rejected.

4.5.2 Corrosion resistance. The blackout marker light shall successfully withstand 200 hours of salt atmosphere in accordance with ASTM B117.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The blackout marker lights covered by this specification are intended for use in various types of ground vehicles.

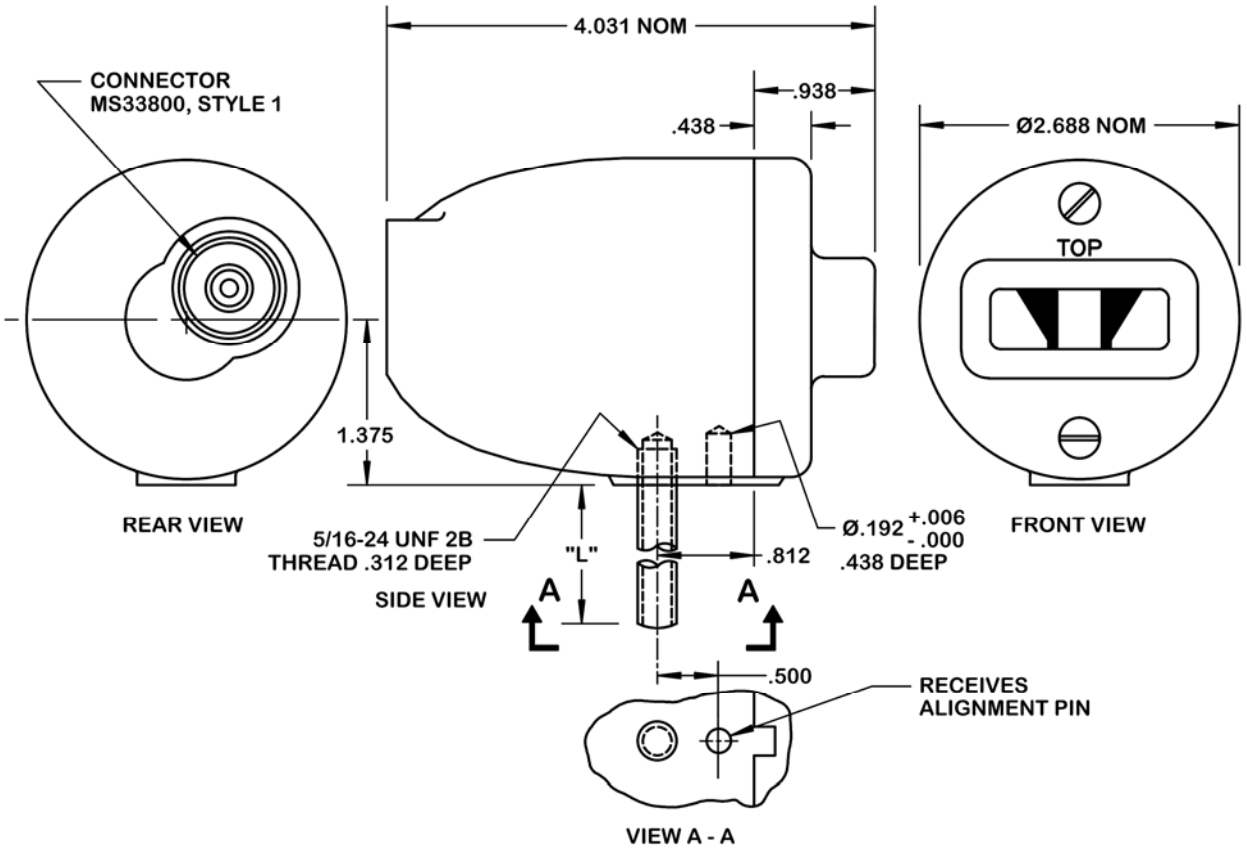
6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type (see 1.2).
c. AQL (see 4.2.2).
d. Conformance inspection (see 4.3).
e. Verification alternatives (see 4.4.1).
f. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Corrosion resistance
Fungus resistance
Lamp
Salt atmosphere
Salt spray
Tactical lighting
Vehicle lighting
Waterproofness

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NOTES:
1. All dimensions are in inches.
2. Dimension L is 1 7/8 inch for MS51303-1, and 7/8 inch for MS51303-2.

FIGURE 1. Light, marker, blackout.

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Preparing Activity:
DLA - GS2

(Project 6220-2010-014)

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